

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal

New Scheme Based On AICTE Flexible Curricula

Electric Vehicles VIII- Semester

EV- 801	Modelling and Simulation of Electric Hybrid Vehicles	2L-1T-2P	4 Credits
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Course Objectives:

1. To understand, model and simulate e-vehicles Motor Drives
2. To understand Energy Storage Systems and effects of working conditions.
3. To analyse the control system, methods and strategies for electric vehicles.
4. To understand electric vehicle configuration for types of electric vehicles.
5. To understand the vehicle dynamics for electric vehicle.

Course Outcomes:

On completion of the course the learner will be able to;

CO1. CLASSIFY and EVALUATE Motor Drives

CO2. MODEL and EVALUATE Energy Storage Systems

CO3. SIMULATE and EVALUATE of ECU, BCU, MCU required for e-Vehicles.

CO4. CATEGORIZE Electric and Hybrid Vehicle Configuration based on Propulsion and Power distribution.

CO5. SIMULATE and DESIGN of Transmission system for e-Vehicles.

Syllabus:

UNIT 1 : Prime Movers [Electric Motor]

Motor Drives for EV (using DC Motor Drives, Induction Motor Drives, Permanent Magnet Brushless and Switched Reluctance Motor Drives), Selection of Motor, Structural Configuration of motor layout (single motor, dual motor, in wheel/Hub motor, Planetary Geared Motors, etc) for EV, Motor Safety and Maintenance, Motor Torque and Power Rating.

UNIT 2: Energy Storage Systems [Battery/Cell Pack]

Types and Packs with respect to Construction, Working, Comparison and Selection (lead-acid, nickel based, lithium-based batteries), Noise Factors, Battery Packs design against Noise and Vibration exposure, Vibration exposure (Mode shapes), Vehicle Dynamics, Battery Pack, Cooling System and Thermal Management.

UNIT 3: Control Units

Introduction-Motor Control System, Modes of Control, Electronic Control Unit [ECU], Battery/Cell Control System, Modes of Control, Battery Control Unit [BCU], Sensor Management and Integration, EV and EHV configuration based on power electronics. Control Methods and Strategies - Torque Coupling and Speed Coupling, Speed and Torque Controllers, BCU, MCU, Speed control for constant torque, constant HP operation of all electric motors [DC/DC chopper, inverter based V/f Operation (motoring and braking - regenerative braking)], Control methods (Phase Flux Linkage, Phase Inductance, Modulated Signal Injection, Mutually Induced Voltage, Observer -Based).

UNIT 4: Electric Vehicle Configuration

Electric Vehicle Configuration, Layouts with respect to Steering, Propulsion and Power distribution - Unicycle, Two-Wheeler (Bicycle, Dicycle, Motorcycle, Scooter, Scooteretts, Mopeds and Underbone), Three-Wheeler, and Four-Wheeler Electric Vehicles, Steering and Propulsion Configuration, Placement of Motors, Battery and Motion Transmission Systems, Drive-Train Topologies.

UNIT 5: Frame/Chassis Design

Introduction to body loads (Load cases and load factor, road loads), Vehicle Dynamics, Vehicle Structure/Chassis/Frames, Strength and Stiffness, Frame building Problems, frame components, Front and Rear Suspension Systems, Battery Packs, Understructure Design, FMEA (Failure Mode and Effects Analysis), Vehicle Structure design against Noise and Vibration exposure, Retrofitting and its associated Problems.

Text Books:

1. Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay, Ali Emadi (2005), "Modern Electric, Hybrid Electric, and Fuel cell vehicles, Fundamentals, Theory, and Design" CRC Press. ISBN 0-8493-3154-4.
2. Chan, C.C. and Chau, K.T., (2001), "Modern Electric Vehicle Technology," Oxford University Press, ISBN: 9780198504160
3. Das, Shuvra, (2021), "Modeling for Hybrid and Electric Vehicles Using Simscape," Springer, ISBN: 9783031003806
4. Du, H., Cao, D., Zhang, H., (2017), "Modeling, Dynamics, and Control of Electrified Vehicles," Woodhead Publishing, ISBN: 9780128127865
5. Scrosati, B., Garche, J., Tillmetz, W., (2015), "Advances in Battery Technologies for Electric Vehicles," Woodhead Publishing, ISBN: 9781782423775
6. Szumanowski, A., (2013), "Hybrid Electric Power Train Engineering and Technology: Modeling, Control, and Simulation," Hybrid Electric Power Train Engineering and Technology: Modeling, Control, and Simulation, ISBN: 9781466640429.
7. Varga, B. O., Mariasiu, F., Moldovanu, D., Iclodean, C., (2016), "Electric and Plug-In Hybrid Vehicles: Advanced Simulation Methodologies," Springer, ISBN: 9783319384443
8. Dincer, I., Halil S. Hamut, H. S., Javani, N., (2017), "Thermal Management of Electric Vehicle Battery Systems," Wiley, ISBN: 9781118900246

9. Wang, S., Fernandez, C., Chunmei, Y., Fan, Y., Wen, C., Stroe, D-I., Chen, Z., (2021), "Battery System Modeling," Elsevier, ISBN: 9780323904728
10. Vangi, D., (2020), "Vehicle Collision Dynamics: Analysis and Reconstruction," Butterworth-Heinemann, ISBN: 9780128127506
11. Gokhale, N. S., (2021), "Practical Fatigue and Durability Analysis," Finite To Infinite, ISBN: 9788195450923

List of Experiments

Student Shall complete minimum 8 practical's from the list given below

1. Estimation of power rating of traction motor for different gradeability by using software viz. MATLAB/ Simulink.
2. Estimation of power rating of traction motor for maximum vehicle speed by using software viz. MATLAB/ Simulink.
3. Introduction to Battery Module and Simulating a Single Battery Cell Using the MSMD / ECM / NTGK Battery Model.
4. Experimental analysis of hybridization of Battery (li-ion / lead acid) and super-capacitor for evaluating their performance parameters.
5. Simulation of application oriented electric vehicles for component sizing and positioning and power consumption of electric vehicles and evaluation of performance characteristics for different drive cycles.
6. Simulation of equivalent circuit modeling for electric vehicle 2RC / 3RC battery.
7. Simulation of EV Power Train by using MATLAB/ Simulink.
8. Structural analysis of single stage transmission systems used in e vehicles.
9. Effect of various performance parameters (speed, gradeability, etc.) on total tractive efforts using MATLAB/ Simulink.
10. Sizing of motor power for different operating conditions for two wheeler/ four wheeler application.
11. Modeling and Simulation of Battery crash analysis using suitable FEA software.
12. Structural Analysis& Modeling of battery packs for different applications (2w/3w/4w).
13. Demonstration on the components of electric scooters.

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Electric Vehicles VIII- Semester

EV -802 (a)	Tribology and Maintenance Engineering	3L-1T-0P	4 Credits
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Course Objectives:

- 1.To develop a solution oriented approach by in depth knowledge of Industrial Tribology.
- 2.To address the underlying concepts, methods and application of Industrial Tribology.

Course Outcomes:

After studying this course, students will be able to;

CO1:Understand fundamental concepts of Contact stresses, residual stress fracture mechanics

CO2: Design friction devices

CO3:Determine wear rate

CO4: Compare SAE,BIS, ASTM, IP, DIN Standards

CO5: Perform Instrumental Tests used in Nano Tribology

Unit-1 : Introduction of Tribology

Introduction, history of tribology, early scientific, Studies of - friction, wear, Lubrication. Tribo-Surface preparations and characteristics. ,Surface contacts, Hertz contact stresses, Residual stress, surface fatigue, creep, Stress relaxation, fracture mechanics, elastic, Viscoelastic and plastic behaviour of materials. Choice of materials

UNIT-2 : Friction

Friction, laws of friction, rolling/sliding friction, Theory of adhesion and abrasion ,Different mechanisms of friction , stick slip characteristics, Interface temperature, thermal analysis, Molecular mechanical theory of friction, Operating conditions and system parameters, Calculations of coefficient of friction, Design of friction devices.

UNIT-3 : Wear

Different types of wear mechanisms, adhesive, Abrasive impact, percussion erosion, Fretting wear calculations of wear rate, two body/ three body wear, Wear prevention, wear of metal cutting and metal forming tools, Wear mapping of materials, cavitation, Surface fatigue, corrosion, Performance levels classifications and specifications of lubricants

UNIT-4 : Lubrication

Lubricants and additives, composition and properties of lubricants, Maintenance of oil and emulsions, industrial hygiene aspects, Technical regulations for lubricants. Boundary/ mixed and fluid film lubrication, industrial methods of lubrications, SAE,BIS, ASTM, IP, DIN Standards, Oil testing's, wear and chemistry of lubricants.

UNIT-5 : Nano Tribology

Instrumental tests,. Bearings, Clutches and brakes, slide units, Dynamic seals, Automobile applications, Machine tools/ press machines applications. Other applications and case studies.

Tutorial topics:

1. Testing equipment of tribology.
2. Various industrial applications of tribology.
3. NEMS and MEMS applications
4. Solid, liquid and mist/ gas lubricants.
5. Surface coatings.
6. Chemical analysis of materials
7. Various simulations
8. AFM/ FFM , SFA, STM, studies.

REFERENCES

1. Fundamentals of Tribology, Basu, SenGupta and Ahuja/PHI
2. Tribology in Industry : Sushil Kumar Srivatsava, S. Chand &Co.
3. Tribology H.G.Phakatkar and R.R.Ghorpade Nirali Publications
4. Tribology – B.C. Majumdar, McGraw Hill Co Ltd.
5. Standard Hand Book of Lubrication Engg., O'Conner and Royle, McGraw Hills C
6. Introduction to Tribology, Halling , Wykeham Publications Ltd.
7. Lubrication, Raymono O. Gunther; Bailey Bros & Swinfan Ltd.
8. Bearing Systems, Principles and Practice, PT Barwill
9. Tribology Hand Book, Michel Ncole

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New Scheme Based On AICTE Flexible Curricula **Electric Vehicles , VIII-Semester**

EV -802 (b)	Rapid Prototyping & Reverse Engineering	3L-1T-0P	4 Credits
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Course Objectives:

After learning the course , students will be able to:

CO 1: Use basic 3D CAD package to produce 3D surfaces and solid models.

CO2: Understand the process of Digital prototyping , Virtual prototyping, Rapid Tooling.

CO 3: Identify various types of rapid prototyping machines.

CO4: Know the basic fundamentals of Reverse Engineering and CAD Modeling

CO5: Prepare CAD Model

Syllabus

UNIT 1: RAPID PROTOTYPING

History and Overview of Rapid Prototyping, General Usage of Rapid Prototyping, Data Translations (CAD to Rapid), File Format (STL - Stereolithography), Operation of Rapid Prototyping Machine,

UNIT 2 : VIRTUAL PROTOTYPING

Need - Development of RP systems – RP process chain - Impact of Rapid Prototyping on Product Development –Digital prototyping - Virtual prototyping- Rapid Tooling - Benefits-Applications.

UNIT 3: REVERSE ENGINEERING AND CAD MODELING

Basic concept- Digitization techniques, Model Reconstruction, Data Processing for Rapid Prototyping: CAD model preparation, Data Requirements, geometric modeling techniques: Wireframe, surface and solid modeling, data formats, Data interfacing, Part orientation and support generation, Support structure design, Model

Slicing and contour data organization, direct and adaptive slicing, Tool path generation.

UNIT 4: LIQUID-BASED AND SOLID-BASED RAPID PROTOTYPING SYSTEMS

Stereolithography (SLA): Apparatus: Principle, per-build process, part-building, post build processes, photopolymerization of SL resins, part quality and process planning, recoating issues, materials, advantages, limitations and applications. Solid Ground Curing (SGC): working principle, process, strengths, weaknesses and applications. Fused deposition Modeling (FDM): Principle, details of processes, process variables, types, products, materials and applications. laminated object manufacturing(LOM): Working Principles, details of processes, products, materials, advantages, limitations and applications - Case studies.

UNIT 5: POWDER-BASED RAPID PROTOTYPING SYSTEMS

Selective Laser Sintering(SLS): Principle, process, Indirect and direct SLS- powder structures, modeling of SLS, materials, post-processing, post-curing, surface deviation and accuracy, Applications. Laser Engineered Net Shaping (LENS): Processes, materials, products, advantages, limitations and applications– Case Studies.

TEXT BOOKS:

1. Rapid prototyping: Principles and applications, second edition, Chua C.K., Leong K.F., and Lim C.S., World Scientific Publishers, 2003.
2. Rapid prototyping, Andreas Gebhardt, Hanser Gardener Publications, 2003.

REFERENCES:

1. Rapid Prototyping and Engineering applications : A tool box for prototype development, Liou W.Liou, Frank W.Liou, CRC Press, 2007.
2. Rapid Prototyping: Theory and practice, Ali K. Kamrani, Emad Abouel Nasr, Springer, 2006.
3. Rapid Tooling: Technologies and Industrial Applications, Peter D.Hilton, Hilton/Jacobs, Paul F.Jacobs, CRC press, 2000.

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EV -802 (c)	Intelligent Manufacturing	3L-1T-0P	4 Credits
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Course Outcomes: After completion of the course, the students will be able to:

- Summarize the concepts of computer integrated manufacturing systems and manufacturing communication systems.
- Identify various components of knowledge based systems.
- Demonstrate the concepts of artificial intelligence and automated process planning.
- Select the manufacturing equipment using knowledge based system for equipment selection.
- Apply various methods to solve group technology problems and demonstrate the structure for knowledge based system for group technology

Syllabus:

Unit-I:

Computer Integrated Manufacturing Systems: Structure and functional areas of CIM system, - CAD, CAPP, CAM, CAQC, ASRS. Advantages of CIM.

Unit-II:

Manufacturing Communication Systems: MAP/TOP, OSI Model, Data Redundancy, Top- down and Bottom-up Approach, Volume of Information. Intelligent Manufacturing System Components, System Architecture and Data Flow, System Operation.

Unit-III:

Basic Components of Knowledge Based Systems: Knowledge Representation, Comparison of Knowledge Representation Schemes, Inference Engine, Knowledge Acquisition. Automated Process Planning - Variant Approach, Generative Approach,

Unit-IV:

Expert Systems for Process Planning: Feature Recognition, Phases of Process planning. Knowledge Based System for Equipment Selection (KBSES) - Manufacturing system design. Equipment Selection Problem, Modeling the Manufacturing Equipment Selection Problem, Problem Solving approach in KBSES, Structure of the KBSES.

Unit-V:

Group Technology: Models and Algorithms Visual Method, Coding Method, Cluster Analysis Method, Matrix Formation - Similarity Coefficient Method, Sorting-based Algorithms, Bond Energy Algorithm, Cost Based method, Cluster Identification Method, Extended CI Method. Knowledge Based Group Technology - Group Technology in Automated Manufacturing System. Structure of Knowledge based system for group technology (KBSC IT) — Data Base, Knowledge Base, Clustering Algorithm.

Suggested Text Books:

- (i) Andrew Kusiak, "Intelligent Manufacturing Systems", Prentice Hall, 1990.
- (ii) Pat Langley, "Computational Intelligence and Intelligent Systems", 2006.

Suggested Reference Books:

- (i) Mohammad Jamshidi, "Design and Implementation of Intelligent Manufacturing Systems: From Expert Systems, Neural Networks to Fuzzy Logic", 1st Edition, 1995.
- (ii) Lucia Knapčiková, Michal Balog, "Industry 4.0: Trends in Management of Intelligent Manufacturing Systems", Springer, 2019.

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EV -803 (a)	IPR (Intellectual Property Rights)	3L-0T-0P	3 Credits
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Course Outcomes:

After studying this course, students will be able to;

1. Aware of their rights for the protection of their invention done in their project work.
2. Know the registration process in our country and foreign countries of their invention, designs and thesis or theory written by them during their project work and for this they must have knowledge of patents, copy right, trademarks, designs and information Technology Act.
3. Identify the different types of IPRs.
4. Know about the registration of Trade Marks, Infringement & Remedies, Digital signature and Electronic Signature. Cybercrimes.
5. Understand about legal provisions in E-Commerce, E-Governance

Syllabus

UNIT 1: INTRODUCTION TO IPR

Meaning of property, Origin, Nature, Meaning of Intellectual Property Rights – 2 hours
Introduction to TRIPS and WTO. Kinds of Intellectual property rights—Copy Right, Patent, Trade Mark, Trade Secret and trade dress, Design, Layout Design, Geographical Indication, Plant Varieties and Traditional Knowledge.

UNIT 2 PATENT RIGHTS AND COPY RIGHTS

Origin, Meaning of Patent, Types, Inventions which are not patentable, Registration Procedure, Rights and Duties of Patentee, Assignment and licence , Restoration of lapsed Patents, Surrender and Revocation of Patents, Infringement, Remedies & Penalties. – 6 hours
COPY RIGHT— Origin, Definition &Types of Copy Right, Registration procedure, Assignment & licence, Terms of Copy Right, Piracy, Infringement, Remedies, Copy rights with special reference to software.

UNIT 3 TRADE MARKS

Origin, Meaning & Nature of Trade Marks, Types, Registration of Trade Marks, Infringement and remedies, Offences relating to Trade Marks, Passing Off, Penalties. Domain Names on cyber space

UNIT 4 DESIGN

Meaning, Definition, Object, Registration of Design, Cancellation of Registration, International convention on design, functions of Design. Semiconductor Integrated circuits and layout design Act-2000.

UNIT 5 BASIC TENENTS OF INFORMATION TECHNOLOGY ACT-2000

IT Act - Introduction E-Commerce and legal provisions E-Governance and legal provisions Digital signature and Electronic Signature. Cybercrimes,

References:

1. Intellectual Property Rights and the Law, Gogia Law Agency, by Dr. G.B. Reddy
2. Law relating to Intellectual Property, Universal Law Publishing Co, by Dr. B.L.Wadehra
3. IPR by P. Narayanan 4. Law of Intellectual Property, Asian Law House, Dr.S.R. Myneni.

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EV -803 (b)	Block Chain Technologies	3L-0T-0P	3 Credits
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COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Understand Nakamoto's concept with Block chain based crypto currency and Technologies Borrowed in Block chain

CO2: Know the concept of Basic Distributed Computing & Crypto primitives

CO3: Explain the philosophy of Bit coin block chain, Challenges and solutions

CO4: Writing smart contracts using Solidity & JavaScript

CO5: Understand the fundamentals of Privacy, Security issues in Block Chain

Syllabus:

UNIT 1: Introduction

Need for Distributed Record Keeping, Modeling faults and adversaries, Byzantine Generals problem, Consensus algorithms and their scalability problems, Nakamoto's concept with Blockchain based cryptocurrency, Technologies Borrowed in Blockchain – hash pointers, consensus, byzantine fault-tolerant distributed computing, digital cash etc.

UNIT 2: Basic Distributed Computing & Crypto primitives:

Atomic Broadcast, Consensus, Byzantine Models of fault tolerance, Hash functions, Puzzle friendly Hash, Collision resistant hash, digital signatures, public key crypto, verifiable random functions, Zero-knowledge systems

UNIT 3 Bitcoin basics

Bitcoin blockchain, Challenges and solutions, proof of work, Proof of stake, alternatives to Bitcoin consensus, Bitcoin scripting language and their use

UNIT 4. Ethereum basics

Ethereum and Smart Contracts, The Turing Completeness of Smart Contract Languages and verification challenges, Using smart contracts to enforce legal contracts, comparing Bitcoin scripting vs. Ethereum Smart Contracts, Writing smart contracts using Solidity & JavaScript

UNIT 5. Privacy, Security issues in Block Chain

Pseudo-anonymity vs. anonymity, Zcash and Zk-SNARKS for anonymity preservation, attacks on Blockchains: Sybil attacks, selfish mining, 51% attacks advent of algorand; Sharding based consensus algorithms to prevent these attacks

Case Studies: Block chain in Financial Service, Supply Chain Management and Government Service

TEXT / REFERENCE BOOKS

1. Mark Gates, “Block chain: Ultimate guide to understanding block chain, bit coin, crypto currencies, smart contracts and the future of money”, Wise Fox Publishing and Mark Gates 2017.
2. Salman Baset, Luc Desrosiers, Nitin Gaur, Petr Novotny, Anthony O'Dowd, Venkatraman Ramakrishna, “Hands-On Block chain with Hyper ledger: Building decentralized applications with Hyperledger Fabric and Composer”, 2018.
3. Bahga, Vijay Madiseti, “Block chain Applications: A Hands-On Approach”, Arshdeep Bahga, Vijay Madiseti publishers 2017

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Electric Vehicles VIII-Semester

EV 803 (c)	ERP and Management Information System	3L-0T-0P	3 Credits
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.Course Outcomes:

The students will understand the MIS concepts its applications, challenges in implementation of ERP system, ERP System Implementation options, and functional modules of ERP

Syllabus

Unit-I: Introduction MIS importance, definition, nature and scope of MIS, Structure and Classification of MIS, Information and Systems Concept, Types of Information, Information systems for competitive advantage.

Unit-II: Business Applications of Information Systems E-Commerce: E-commerce features & Business Models - Decision Support Systems - Business Process Reengineering - Business Intelligence and Knowledge Management System.

Unit-III: Management of Information Systems Information system planning, system acquisition, systems implementation, evaluation & maintenance of IS, IS Security and Control. Global perspective on cybercrime - Cybercrime era.

Unit-IV: Introduction to ERP ERP System: Overview of ERP Systems, Business benefits of ERP, Vendor Analysis, Challenges of implementing ERP Systems - ERP Maintenance - Emerging Trends in ERP

Unit-V: ERP - Modules Modules: Business Modules in an ERP Package - Manufacturing, Human Resources, Plant Maintenance, Materials Management, and Supply chain Management (SCM), Sales and Distribution. Case Study on Banking Sector

REFERENCES:

1. D P Goyal, Management Information Systems–Managerial Perspective, MacMillan.
2. Laudon & Laudon, Management Information Systems, Pearson. Jawadekar, MIS Text and Cases, TMH.
3. Mary Sumner —Enterprise Resource Planning‖ Pearson.
4. Ellen Monk —Enterprise Resource Planning‖ Cengage.
5. Goyal —Enterprise Resource Planning‖ TMH.

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Electric Vehicles VIII-Semester

EV -804	EHV Lab	0L-0T-6P	3 Credits
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Course Objectives:

To have hands-on experience of using basic e-vehicle technologies and their advancements.

Course Outcomes: On completion of the course, learner will be able to

CO1. RECOGNIZE different e-vehicle technologies

CO2. EVALUATE battery types and capacities

CO3. COMPARE cell balancing

CO4. DEMONSTRATE various power connection types for motor control and battery discharge circuits

CO5. EVALUATE application of specific battery packs CO6. APPRAISE the practically available e vehicle systems

Suggested List of Experiments

1. Study of basic components of e-vehicles.
2. Study of basic components of hybrid vehicles.
3. Battery capacity calculations for specific application.
4. Study and verification active and passive cell balancing (using suitable simulation).
5. Battery connections for discharge system (using suitable simulation).
6. Experiment/Simulation for AC-DC, DC-DC, Speed Control using electric motor.
7. Battery pack performance characteristics (To know the variation of time with various battery working parameters).
8. Determination of suitable wire size for specific capacity of motor.
9. Study of different wire harnessing for e-vehicle.
10. Study of Battery Management System.
11. Case study of 2/3/4 wheeler e-vehicle/hybrid vehicle

Evaluation: Evaluation will be continuous and integral part of the class as well as through external assessment (Viva/voce)

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Electric Vehicles VIII-Semester

EV-805	Major Project -II	0L-0T-8P	3 Credits
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Objectives of the course Major Project II:

- To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- To give students an opportunity to do some thing creative and to assimilate real life work situation in institution.
- To adapt students for latest development and to handle independently new situations.
- To develop good expressions power and presentation abilities in students.

The focus of the Major Project II (which may be extension of Major Project I or New topic of project) is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system (if any)

Working schedule:

The faculty and students should work according to following schedule: Each student undertakes substantial and individual project in an approved area of the subject and supervised by a faculty of the department. In special case, if project is huge, then maximum 03 students may be permitted to work together as a team to do the same. The student must submit outline and action plan for the project execution (time schedule) and the same be approved by the concerned faculty and Head of department.

Project guide should motivate students to develop some Innovative working models in the emerging area of Electric Vehicles, EHV, Advanced Automotives, Aero Modelling, Advanced Manufacturing Technology based systems etc. which can contribute to the society.

Evaluation: There will be both external and internal evaluation of project carried out by each student.
